NAME: _____ Geometry (FL B.E.S.T.)

9.4 Circle Equations

A circle is the set of points a fixed distance r from a center (h, k). Applying the distance formula to a general point (x, y) on the circle gives $\sqrt{(x-h)^2+(y-k)^2} = r$, and squaring both sides produces the **standard form**: $(x-h)^2+(y-k)^2 = r^2$. If an equation is given in expanded form ($x^2+y^2+Dx+Ey+F=0$), **completing the square** on the x-terms and the y-terms separately converts it back to standard form, revealing the center and radius. Once you have standard form, you can test whether a point is inside, on, or outside the circle by comparing $(x-h)^2+(y-k)^2$ to r^2: less than means inside, equal means on, greater means outside. This worksheet also finds where a circle crosses the axes and includes a coverage-radius application.

Practice Set — All Questions

15 questions. Work them on your own paper, then check the key below.

1. Center $(2, -5)$, radius 4. Write the equation.

2. Find center and radius of $(x+3)^2+(y-7)^2 = 36$.

3. Center $(0, 0)$ through $(3, 4)$. Write the equation.

4. Rewrite $x^2+y^2-4x+6y-3=0$ in standard form.

5. Is $(5, 3)$ inside, on, or outside $(x-2)^2+(y+1)^2 = 36$?

6. Center $(-1, -1)$, radius 2. Write the equation.

7. Find center and radius of $x^2+(y-3)^2 = 1$.

8. Center $(1, 2)$ through $(4, 6)$. Write the equation.

9. Rewrite $x^2+y^2+2x-10y+17=0$ in standard form.

10. Is $(0, 0)$ on $x^2+y^2 = 1$?

11. What is r for $(x-5)^2+(y+4)^2 = 64$?

12. What is the center of $(x+9)^2+y^2 = 4$?

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13. Is $(3, 4)$ on $x^2 + y^2 = 25$?

14. Find where circle $(x - 4)^2 + (y - 1)^2 = 25$ crosses the y-axis.

15. Rewrite $x^2 + y^2 - 10x - 4y + 13 = 0$ in standard form, then determine whether $(7, 6)$ is inside, on, or outside the circle.

✓ Answer Key

1. $(x - 2)^2 + (y + 5)^2 = 16$
2. Center $(-3, 7)$, radius 6
3. $x^2 + y^2 = 25$
4. $(x - 2)^2 + (y + 3)^2 = 16$
5. Inside $(25 < 36)$
6. $(x + 1)^2 + (y + 1)^2 = 4$
7. Center $(0, 3)$, radius 1
8. $(x - 1)^2 + (y - 2)^2 = 25$
9. $(x + 1)^2 + (y - 5)^2 = 9$
10. No — inside $(0 < 1)$
11. 8
12. $(-9, 0)$
13. Yes (on the circle)
14. $(0, 4)$ and $(0, -2)$
15. $(x - 5)^2 + (y - 2)^2 = 16$; $(7, 6)$ is outside $(20 > 16)$